Channel representations of noisy coherent detection in continuous-variable quantum key distribution: Additive-noise versus attenuator

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(Dated: August 18, 2026)

We study the security of continuous-variable quantum key distribution (CV-QKD) with imperfect coherent receivers when the receiver noise is treated as untrusted. For each detection scheme, we construct additive-noise and attenuator channels that, together with a rank-one reference POVM and an invertible deterministic rescaling of the outcomes, reproduce the same noisy POVM. We derive the channel parameters and compare the resulting Holevo information and asymptotic secret fraction.

For the parameter sets considered, the receiver imperfections reduce the asymptotic secret fraction relative to ideal detection for both channel choices. The channel giving the lower secret fraction depends on the receiver parameters and the communication-channel transmittance. For homodyne detection, the attenuator channel gives a lower secret fraction at high transmittance. Its secret fraction remains positive over a longer distance.

For double-homodyne detection, the channel parameters, Holevo information, and asymptotic secret fraction depend on the squeezing parameter r that specifies the reference POVM. The Alice-Bob mutual information is independent of the channel choice and r because all admissible choices reproduce the same receiver statistics. For the parameter sets considered, the Holevo information is concave in r for the additive-noise channel and convex in r for the attenuator channel. The channel choice and the value of r therefore affect the asymptotic secret fraction.

I. INTRODUCTION

Homodyne and double-homodyne detection are standard coherent detection schemes in quantum optics and continuous-variable quantum information [1, 2]. Here, double-homodyne detection denotes the two-branch receiver conventionally used to implement heterodyne measurement. In the limit of a strong local oscillator, an ideal homodyne receiver implements a projective measurement of one field quadrature. An ideal balanced double-homodyne receiver implements a joint noisy measurement of two conjugate quadratures described by the coherent-state positive operator-valued measure (POVM), with POVM density proportional to $|\alpha\rangle\langle\alpha|$ [2, 3]. Both schemes are Gaussian measurements and produce Gaussian outcome distributions for Gaussian input states. They are used in quantum state tomography [4, 5], quantum state preparation [6], and continuous-variable quantum key distribution (CV-QKD) [7–9].

Practical receivers may have detector quantum efficiencies below unity, detector-efficiency mismatch, and beam-splitter imbalance. Within the Gaussian approximation to the photon-count statistics in the strong local oscillator limit, these receiver imperfections produce noisy Gaussian POVMs [10–12]. They also affect CV-QKD parameter estimation and the asymptotic secret fraction [13–15]. We use *receiver imperfections* for all these deviations from ideal detection and reserve *receiver asymmetry* for beam-splitter imbalance and detector-efficiency mismatch; Ref. [12] refers to this combination as *asymmetry effects*. For each detection scheme, the receiver parameters determine a noisy Gaussian POVM, which we call the target POVM. We denote the sender, receiver, and eavesdropper by Alice, Bob, and Eve, respectively.

General quantum measurement theory separates the outcome statistics from the measurement-induced state changes. A POVM determines the outcome probabilities. A quantum instrument also specifies the conditional state changes. Davies and Lewis introduced instruments to describe general measurements and successive-measurement statistics [16]. Holevo showed that every POVM admits a statistically equivalent projection-valued measurement

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on an extended system containing an auxiliary system in a fixed state; such Naimark extensions are generally nonunique [17]. Ozawa subsequently established that physically realizable measurements are described by completely positive instruments and admit indirect measurement models [18]. A target POVM therefore fixes the outcome probabilities.

Measurement noise can enter through classical processing of the outcomes or through a quantum channel preceding the measurement. Ali and Emch described finite resolution by smearing a sharp observable with a resolution function, which yields a fuzzy observable represented by a POVM [19]. Martens and de Muynck later formalized measurement nonideality through a partial order on POVMs [20]. Quantum preprocessing applies a channel to the input state; equivalently, the dual channel acts on the POVM effects. The corresponding preprocessing order was studied in Ref. [21]. The induced POVM fixes the outcome statistics for every input state. Several preprocessing channels and channel dilations may induce the same POVM.

In coherent optical detection, Yuen and Chan calculated the mean-square fluctuation spectra of homodyne and heterodyne receivers for arbitrary signal and local-oscillator states and separated the contributions of signal quantum fluctuations and local-oscillator excess noise [22]. Holevo described bosonic communication channels through coupling to an auxiliary bosonic system and introduced the linear attenuator as a channel model for signal attenuation [23]. Later classifications place the attenuator and additive-noise channels in different canonical classes of one-mode Gaussian channels [24, 25].

We reserve *communication channel* for the physical Alice-to-Bob link. We use *receiver channel* for an effective one-mode Gaussian channel acting on Bob's mode after the communication channel and before a rank-one reference Gaussian POVM. Both channels considered here admit the same X, Y parametrization:

$$\boldsymbol{r} \mapsto X\boldsymbol{r}, \qquad \Sigma \mapsto X\Sigma X^{\mathrm{T}} + Y, \tag{1}$$

$$X_\Phi = \mathbb{1}, \qquad Y_\Phi \succeq 0, \tag{2}$$

$$X_\mathcal{E} = \cos\theta\,\mathbb{1}, \qquad Y_\mathcal{E} = \sin^2\theta\,\Sigma_{\mathrm{env}}. \tag{3}$$

The additive-noise channel has unit gain. Applied to Bob's mode, it preserves the first moments and the Alice-Bob correlation block while increasing Bob's local covariance by Y_Φ. The attenuator scales the first moments and the Alice-Bob correlation block by $\cos\theta$; its environmental input contributes $\sin^2\theta\,\Sigma_{\mathrm{env}}$ to Bob's local covariance.

After the reference POVM is measured, an invertible deterministic rescaling of the outcomes compensates the factor $\cos\theta$ in the classical first moments. This rescaling leaves the output quantum state unchanged. Suitable choices of Y_Φ, θ, and Σ_{env} allow the two channels to reproduce the same target POVM for every input state. Here, a *channel representation* of the target POVM means the channel, the rank-one reference Gaussian POVM measured at its output, and the invertible deterministic rescaling of the outcomes.

Our previous work derived the target POVMs for asymmetric homodyne and double-homodyne receivers and reproduced them using additive-noise channels [12]. For double-homodyne detection, the reference POVM belongs to a family of rank-one Gaussian POVMs whose effects are proportional to projectors onto displaced squeezed states. The squeezing parameter r labels this family, and $r = 0$ gives the coherent-state POVM of conventional double-homodyne detection. Complete positivity of the additive-noise channel restricts r to an interval determined by the receiver parameters. In the security analysis of Ref. [12], Eve was assumed to purify the Alice-Bob state after the receiver channel. The resulting Holevo information depends on r. A fixed target POVM can therefore be reproduced by several choices of channel and reference POVM that give the same receiver statistics and different Alice-Bob states.

In this paper, we construct attenuator channels for the same target homodyne and double-homodyne POVMs and compare them with the corresponding additive-noise channels. For each detection scheme, we choose the channel parameters and the reference POVM so that the dual map, followed by an invertible deterministic rescaling of the outcomes, reproduces the target POVM. This equality holds for every input state. The Alice-Bob mutual information is therefore independent of the channel choice and, for double-homodyne detection, the value of r. We compare the resulting Holevo information and asymptotic secret fraction for the Gaussian-modulated coherent-state (GMCS) protocol with reverse reconciliation under collective Gaussian attacks [7, 26, 27].

The security interpretation depends on the trust assigned to the receiver noise. In a model with trusted receiver noise, Eve has no access to the environmental output of the receiver [28–31]. For a fixed Alice-Bob-Eve state before that channel, any two choices that reproduce the same target POVM produce the same joint classical-quantum state of Eve and Bob's rescaled outcome. They therefore give the same Holevo information. In the model with untrusted receiver noise used here, Eve purifies the Alice-Bob state after the receiver channel. The additive-noise and attenuator channels generally produce different Alice-Bob states and hence different Holevo information and asymptotic secret fractions. Selecting the channel therefore specifies a physical assumption in the security analysis.

This paper is organized as follows. Section II defines the target homodyne and double-homodyne POVMs. Section III constructs the corresponding additive-noise and attenuator channels. Section IV evaluates the Alice-Bob mutual information, Holevo information, and asymptotic secret fraction. Section V discusses the results.

II. TARGET GAUSSIAN POVMS

We now specify the target homodyne and double-homodyne POVMs produced by the receiver imperfections considered in this work, as shown in Fig. 1. Ref. [12] gives the detailed derivations used in this section.

For the homodyne receiver, $\hat{a}$ is the annihilation operator of the signal mode, α and α_L are the coherent amplitudes of the signal and local oscillator (LO), and C and S are real beam-splitter amplitude coefficients satisfying $C^2+S^2=1$. The photodetectors have quantum efficiencies η_1 and η_2, and $\mu = m_1 - m_2$ is their photon-count difference.

The Q symbol of the target homodyne POVM (see Fig. 1a) is

$$\mathsf{P}_G^{(\mathrm{H})}(x) = \frac{1}{\sqrt{2\pi\sigma_x}} \exp\left[-\frac{(x-\langle\hat{x}_\phi\rangle)^2}{2\sigma_x}\right], \tag{4}$$

$$\langle\hat{x}_\phi\rangle \equiv \langle\alpha|\hat{x}_\phi|\alpha\rangle = 2\,\mathrm{Re}(\alpha e^{-i\phi}), \quad \phi = \arg\alpha_L, \tag{5}$$

$$\hat{x}_\phi = \hat{a}e^{-i\phi} + \hat{a}^\dagger e^{i\phi}, \tag{6}$$

where $\hat{x}_\phi$ is the phase-rotated quadrature operator. The rescaled quadrature variable is

$$x \equiv \frac{\mu}{(\eta_1+\eta_2)CS\,|\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1+\eta_2)\,CS}\,|\alpha_L|\,, \tag{7}$$

and its variance is

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1+\eta_2)CS]^2}. \tag{8}$$

Taking the inverse Fourier transform of the Q symbol (4) gives the antinormally ordered characteristic function and determines the target covariance matrix $\Sigma_m^{(\mathrm{H})}$ [32]. The reference homodyne POVM is the ideal projective quadrature measurement. We retain the subscript id used in Ref. [12] to denote the noiseless reference POVMs. We use an auxiliary squeezing parameter s to write the homodyne measurement as an infinite-squeezing limit. The target and reference covariance matrices are

$$\Sigma_m^{(\mathrm{H})} = \lim_{s\to+\infty} R(\phi)\ \mathrm{diag}(e^{-2s}+\sigma_N, e^{2s})\,R^{\mathrm{T}}(\phi), \tag{9}$$

$$\Sigma_{\mathrm{id}}^{(\mathrm{H})} = \lim_{s\to+\infty} R(\phi)\ \mathrm{diag}(e^{-2s}, e^{2s})\,R^{\mathrm{T}}(\phi), \tag{10}$$

where

$$R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \tag{11}$$

is the rotation matrix and σ_N is the receiver excess-noise variance,

$$\sigma_N = \sigma_x - 1. \tag{12}$$

For double-homodyne detection, (C_S, S_S) and (C_L, S_L) are the real amplitude coefficients of the signal and LO beam splitters, while (C_i, S_i) are those of homodyne branch $i \in \{1,2\}$. Each pair obeys $C_j^2 + S_j^2 = 1$ for $j \in \{S, L, 1, 2\}$. The quantities $\eta_k^{(i)}$ and $m_k^{(i)}$ are the quantum efficiency and photon count of detector $k \in \{1,2\}$ in branch i, and $\mu_i = m_1^{(i)} - m_2^{(i)}$.

The Q symbol of the target double-homodyne POVM is

$$\mathsf{P}_G^{(\mathrm{DH})}(x_1,x_2) = \frac{1}{\pi\sqrt{\sigma_1\sigma_2}} \exp\left[-\frac{[x_1-\mathrm{Re}(\alpha e^{-i\phi})]^2}{\sigma_1} - \frac{[x_2-\mathrm{Im}(\alpha e^{-i\phi})]^2}{\sigma_2}\right], \tag{13}$$

where $\alpha_L^{(1)} = C_L\alpha_L$ and $\alpha_L^{(2)} = -iS_L\alpha_L$ are the local-oscillator amplitudes in the two homodyne branches, and x_i are the rescaled quadrature variables:

$$x_1 = \frac{1}{2(\eta_1^{(1)}+\eta_2^{(1)})C_1S_1C_S}\left\{\frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)}S_1^2 - \eta_2^{(1)}C_1^2)|\alpha_L^{(1)}|\right\}, \tag{14}$$

$$x_2 = \frac{1}{2(\eta_1^{(2)}+\eta_2^{(2)})C_2S_2S_S}\left\{\frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)}S_2^2 - \eta_2^{(2)}C_2^2)|\alpha_L^{(2)}|\right\}. \tag{15}$$

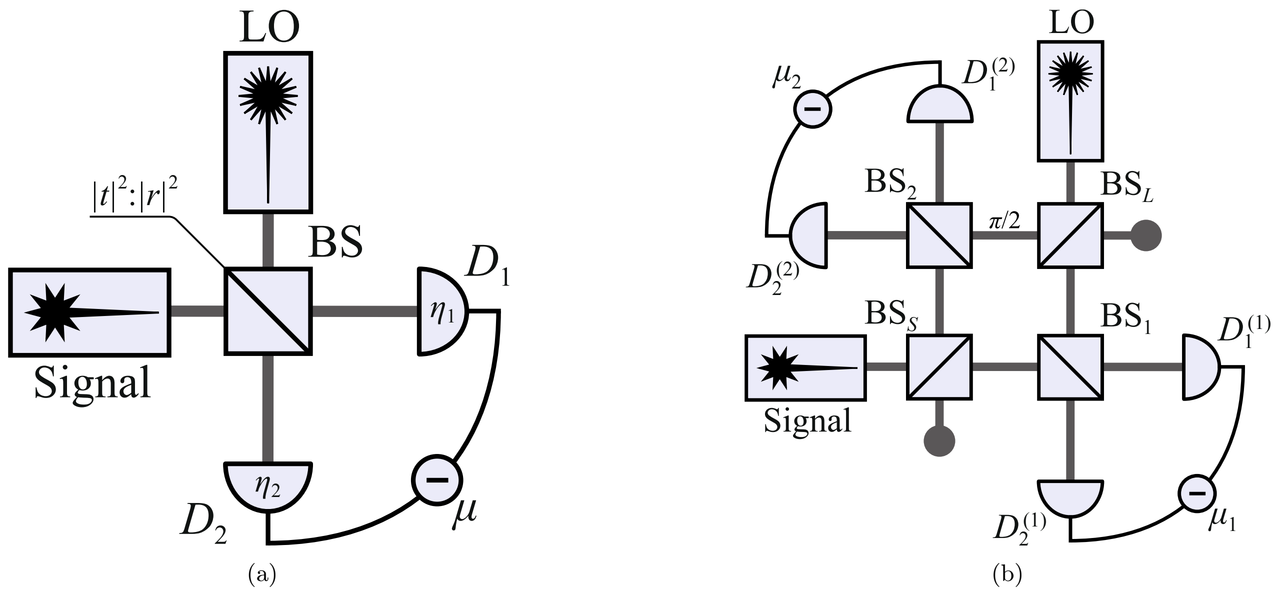


Figure 1: (a) Scheme of homodyne receiver. S (LO) is the source of signal (reference) mode, BS is the beam splitter with transmission and reflection coefficients C and S, respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2, and $\mu = m_1 - m_2$ is the photon count difference. (b) Scheme of double-homodyne receiver. S (LO) is the source of signal (reference) mode, BS_S (BS_L) is the signal mode (local oscillator) beam splitter; $\frac{\pi}{2}$ is the quarter wave phase shifter; BS_i is the beam splitter of ith homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the ith homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of ith homodyne.

The parameters σ_i are defined by

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \tag{16}$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2}{\left[(\eta_1^{(i)} + \eta_2^{(i)}) C_i S_i\right]^2}. \tag{17}$$

Thus, for fixed input amplitude α, the conditional outcome variances are $\mathrm{Var}(x_i \mid \alpha) = \sigma_i/2$.

Following the same procedure used to derive Eq. (9), we obtain the covariance matrix of the target double-homodyne POVM,

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(\delta_1, \delta_2) R^{\mathrm{T}}(\phi), \tag{18}$$

$$\delta_i = 2\sigma_i - 1. \tag{19}$$

For a complex outcome $z = z_1 + iz_2$, we define the effects of the rank-one reference Gaussian POVM by

$$\hat{\Pi}_{\mathrm{id}}^{(z)} = \frac{1}{\pi} |z, r, \phi\rangle\langle z, r, \phi|, \qquad z_1, z_2 \in \mathbb{R}, \tag{20}$$

where

$$|z, r, \phi\rangle = \hat{R}(\phi)\hat{D}(z)\hat{S}(r)|0\rangle = \hat{D}(ze^{i\phi})\hat{S}(re^{2i\phi})|0\rangle, \tag{21}$$

and $r \in \mathbb{R}$ is the squeezing parameter. The operators are $\hat{R}(\phi) = e^{i\phi\hat{a}^\dagger\hat{a}}$, $\hat{D}(z) = e^{z\hat{a}^\dagger - z^*\hat{a}}$, and $\hat{S}(\zeta) = e^{[\zeta\hat{a}^{\dagger 2} - \zeta^*\hat{a}^2]/2}$ for $\zeta \in \mathbb{C}$. At $r = 0$, the reference Gaussian POVM reduces to the coherent-state POVM implemented by conventional ideal double-homodyne detection.

For fixed r and ϕ, the displaced squeezed states in Eq. (21) have covariance matrix

$$\Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^{\mathrm{T}}(\phi). \tag{22}$$

Complete positivity of the additive-noise channel considered below requires $\Sigma_m^{(\mathrm{DH})} - \Sigma_{\mathrm{id}}^{(\mathrm{DH})} \succeq 0$. Equations (18) and (22) therefore restrict r to the interval fixed by the receiver parameters [12],

$$r \in [r_2, r_1]\,,\ r_1 = \ln \delta_1^{1/2},\ r_2 = \ln \delta_2^{-1/2}. \tag{23}$$

III. GAUSSIAN CHANNELS FOR THE TARGET POVMS

Throughout this section, we consider Gaussian channels with zero displacement and zero-mean noise and use units in which the vacuum covariance matrix is $\mathbb{1}$. For measurement type $\gamma \in \{\mathrm{H}, \mathrm{DH}\}$, where H and DH denote homodyne and double-homodyne detection, respectively, let $\mathcal{C}_\gamma$ denote the effective one-mode Gaussian channel, hereafter the receiver channel.

A Gaussian channel $\mathcal{C}_\gamma$ is characterized by real matrices $X_{\mathcal{C}}^{(\gamma)}$ and $Y_{\mathcal{C}}^{(\gamma)}$. It transforms the first-moment vector $\boldsymbol{r}_G$ and covariance matrix Σ_G of a Gaussian state as [2]

$$\begin{aligned} \boldsymbol{r}_G &\mapsto X_{\mathcal{C}}^{(\gamma)} \boldsymbol{r}_G, \\ \Sigma_G &\mapsto X_{\mathcal{C}}^{(\gamma)} \Sigma_G \left[X_{\mathcal{C}}^{(\gamma)}\right]^{\mathrm{T}} + Y_{\mathcal{C}}^{(\gamma)}. \end{aligned} \tag{24}$$

For an arbitrary state $\hat{\rho}$ and POVM density $\hat{\Pi}^{(\boldsymbol{u})}$, the outcome probability density is [33]

$$p(\boldsymbol{u}|\hat{\rho}) = \mathrm{Tr}\left[\hat{\rho}\,\hat{\Pi}^{(\boldsymbol{u})}\right]. \tag{25}$$

We reproduce the target POVM by applying $\mathcal{C}_\gamma$ to the input state before measuring the reference Gaussian POVM. Equivalently, the dual map $\mathcal{C}_\gamma^*$ acts on the reference POVM:

$$\begin{aligned} p_{\mathcal{C}_\gamma}(\boldsymbol{u}|\hat{\rho}) &= \mathrm{Tr}\left[\mathcal{C}_\gamma(\hat{\rho})\,\hat{\Pi}_{\mathrm{id}}^{(\boldsymbol{u})}\right] \\ &= \mathrm{Tr}\left[\hat{\rho}\,\mathcal{C}_\gamma^*\left(\hat{\Pi}_{\mathrm{id}}^{(\boldsymbol{u})}\right)\right]. \end{aligned} \tag{26}$$

For the channels considered below, we write

$$X_{\mathcal{C}}^{(\gamma)} = \kappa_{\mathcal{C}}^{(\gamma)} \mathbb{1}, \qquad \kappa_{\Phi}^{(\gamma)} = 1, \qquad \kappa_{\mathcal{E}}^{(\gamma)} = \cos\theta_\gamma. \tag{27}$$

Let $d_{\mathrm{H}} = 1$ and $d_{\mathrm{DH}} = 2$ denote the number of measured quadratures. We rescale the outcome of the reference POVM according to

$$\boldsymbol{x} = \frac{\boldsymbol{u}}{\kappa_{\mathcal{C}}^{(\gamma)}}. \tag{28}$$

For double-homodyne detection, $\boldsymbol{u} = (z_1, z_2)^{\mathrm{T}}$, so that

$$z = \kappa_{\mathcal{C}}^{(\mathrm{DH})}(x_1 + i x_2). \tag{29}$$

The resulting POVM density with respect to $\mathrm{d}^{d_\gamma}\boldsymbol{x}$ is

$$\hat{\Pi}_{\mathcal{C}_\gamma}^{(\boldsymbol{x})} = \left[\kappa_{\mathcal{C}}^{(\gamma)}\right]^{d_\gamma} \mathcal{C}_\gamma^*\left(\hat{\Pi}_{\mathrm{id}}^{\left(\kappa_{\mathcal{C}}^{(\gamma)}\boldsymbol{x}\right)}\right). \tag{30}$$

The prefactor in Eq. (30) is the Jacobian of the outcome rescaling and preserves the POVM normalization. A channel representation reproduces the target POVM if

$$\hat{\Pi}_{\mathcal{C}_\gamma}^{(\boldsymbol{x})} = \hat{\Pi}_m^{(\boldsymbol{x})} \tag{31}$$

for every outcome $\boldsymbol{x}$.

At the covariance-matrix level, this condition becomes

$$\Sigma_m^{(\gamma)} = \left[\kappa_{\mathcal{C}}^{(\gamma)}\right]^{-2}\left(\Sigma_{\mathrm{id}}^{(\gamma)} + Y_{\mathcal{C}}^{(\gamma)}\right). \tag{32}$$

Because the channels have zero displacement and $X_{\mathcal{C}}^{(\gamma)}$ is proportional to the identity, the outcome rescaling restores the input-dependent first moments. Equation (32), together with the Jacobian in Eq. (30), therefore establishes equality with the normalized target Gaussian POVM.

A. Additive-noise channel

First, we consider the additive-noise channel Φ_γ, characterized by the matrices [32]

$$X_\Phi^{(\gamma)} = \mathbb{1}, \quad Y_\Phi^{(\gamma)} \succeq 0, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{33}$$

The additive-noise channel is self-dual, $\Phi_\gamma^* = \Phi_\gamma$. It leaves the first moment unchanged and transforms the covariance matrix as

$$\Phi_\gamma : \Sigma \mapsto \Sigma + Y_\Phi^{(\gamma)}. \tag{34}$$

Equation (32) requires

$$\Phi_\gamma : \quad \Sigma_{\mathrm{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \Sigma_{\mathrm{id}}^{(\gamma)} + Y_\Phi^{(\gamma)}, \tag{35}$$

and therefore

$$Y_\Phi^{(\gamma)} = \Sigma_m^{(\gamma)} - \Sigma_{\mathrm{id}}^{(\gamma)} \equiv \Sigma_N^{(\gamma)}, \tag{36}$$

where $\Sigma_N^{(\gamma)}$ is the receiver excess-noise covariance matrix.

For homodyne detection, Eqs. (10) and (9) give

$$\Sigma_N^{(\mathrm{H})} = \Sigma_m^{(\mathrm{H})} - \Sigma_{\mathrm{id}}^{(\mathrm{H})} = R(\phi)\,\mathrm{diag}(\sigma_N, 0) R^{\mathrm{T}}(\phi) \succeq 0, \tag{37}$$

where σ_N is given by Eq. (12).

For double-homodyne detection, Eqs. (18) and (22) give

$$\Sigma_N^{(\mathrm{DH})}(r) = \Sigma_m^{(\mathrm{DH})} - \Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi) \begin{pmatrix} \sigma_N^{(1)}(r) & 0 \\ 0 & \sigma_N^{(2)}(r) \end{pmatrix} R^{\mathrm{T}}(\phi) \succeq 0, \tag{38}$$

$$\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad \sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \tag{39}$$

The squeezing parameter r determines the entries of $\Sigma_N^{(\mathrm{DH})}$. For every $r \in [r_2, r_1]$, the excess-noise covariance matrix is positive semidefinite, $\Sigma_N^{(\mathrm{DH})}(r) \succeq 0$.

B. Attenuator channel

The attenuator channel $\mathcal{E}_\gamma$ is characterized by the matrices [32]

$$X_{\mathcal{E}}^{(\gamma)} = \cos\theta_\gamma \mathbb{1}, \quad Y_{\mathcal{E}}^{(\gamma)} = \sin^2\theta_\gamma \Sigma_{\mathrm{env}}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, \tag{40}$$

where $\Sigma_{\mathrm{env}}^{(\gamma)}$ is the covariance matrix of the environmental input state.

Using Eqs. (24) and (32), we choose $\mathcal{E}_\gamma^*$ so that it maps the reference Gaussian POVM to the target POVM. In the rescaled outcome coordinates, its action on the covariance matrix is

$$\mathcal{E}_\gamma^* : \quad \Sigma_{\mathrm{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2\theta_\gamma} \Sigma_{\mathrm{id}}^{(\gamma)} + \tan^2\theta_\gamma \Sigma_{\mathrm{env}}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{41}$$

For homodyne detection, we take the environmental input state to be the vacuum, with covariance matrix

$$\Sigma_{\mathrm{env}}^{(\mathrm{H})} = \mathbb{1}. \tag{42}$$

The attenuator transforms the measured quadrature according to

$$\hat{x}_{\phi,\mathrm{out}} = \cos\theta_{\mathrm{H}} \hat{x}_{\phi,\mathrm{in}} + \sin\theta_{\mathrm{H}} \hat{x}_{\phi,\mathrm{env}}. \tag{43}$$

Let x_{raw} denote the outcome associated with $\hat{x}_{\phi,\mathrm{out}}$. The rescaling $x = x_{\mathrm{raw}}/\cos\theta_{\mathrm{H}}$ corresponds to the rescaled observable

$$\frac{\hat{x}_{\phi,\mathrm{out}}}{\cos\theta_{\mathrm{H}}} = \hat{x}_{\phi,\mathrm{in}} + \tan\theta_{\mathrm{H}} \hat{x}_{\phi,\mathrm{env}}. \tag{44}$$

Since the vacuum quadrature variance is one, the added variance is $\tan^2\theta_{\mathrm{H}}$. Exact equality with the target homodyne POVM therefore requires

$$\tan^2\theta_{\mathrm{H}} = \sigma_N. \tag{45}$$

Since $\sigma_x = 1+\sigma_N$ by Eq. (12), Eq. (45) gives

$$\cos^2\theta_{\mathrm{H}} = \frac{1}{\sigma_x}, \quad \sin^2\theta_{\mathrm{H}} = \frac{\sigma_N}{\sigma_x}. \tag{46}$$

For double-homodyne detection, first suppose that $r_2 < r_1$, equivalently $\delta_1\delta_2 > 1$, and take $r \in (r_2, r_1)$. In the attenuator channel considered here, we take the environmental input state to be a centered pure Gaussian state with covariance matrix

$$\Sigma_{\mathrm{env}}^{(\mathrm{DH})}(r) = R(\phi)\ \mathrm{diag}\left(a(r), a(r)^{-1}\right)\ R^{\mathrm{T}}(\phi), \qquad a(r) > 0. \tag{47}$$

Equations (41), (22), and (18) give

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(\delta_1,\delta_2)R^{\mathrm{T}}(\phi) = R(\phi)\left\{\frac{1}{\cos^2\theta_{\mathrm{DH}}(r)}\ \mathrm{diag}(e^{2r}, e^{-2r}) + \tan^2\theta_{\mathrm{DH}}(r)\ \mathrm{diag}\left(a(r), a(r)^{-1}\right)\right\} R^{\mathrm{T}}(\phi), \tag{48}$$

$$\begin{cases} \delta_1 = \dfrac{e^{2r}}{\cos^2\theta_{\mathrm{DH}}(r)} + \tan^2\theta_{\mathrm{DH}}(r)a(r), \\ \delta_2 = \dfrac{e^{-2r}}{\cos^2\theta_{\mathrm{DH}}(r)} + \tan^2\theta_{\mathrm{DH}}(r)a(r)^{-1} \end{cases}. \tag{49}$$

Solving the system yields

$$\cos^2\theta_{\mathrm{DH}}(r) = \frac{e^{2r}+a(r)}{\delta_1 + a(r)} = \frac{\delta_1 e^{-2r} + \delta_2 e^{2r} - 2}{\delta_1\delta_2 - 1}, \tag{50}$$

$$a(r) = \frac{\delta_1 - e^{2r}}{\delta_2 e^{2r} - 1}. \tag{51}$$

As $r \to r_2^+$ or $r \to r_1^-$, respectively, $a(r)$ or $a(r)^{-1}$ diverges while $\tan^2\theta_{\mathrm{DH}}(r) \to 0$. The products $\tan^2\theta_{\mathrm{DH}}(r)a(r)$ and $\tan^2\theta_{\mathrm{DH}}(r)a(r)^{-1}$ approach finite limits. The environmental covariance matrix has no finite endpoint limit, while the channel matrices satisfy

$$\lim_{r\to r_2^+} X_{\mathcal{E}}^{(\mathrm{DH})}(r) = \mathbb{1}, \qquad \lim_{r\to r_2^+} Y_{\mathcal{E}}^{(\mathrm{DH})}(r) = \Sigma_N^{(\mathrm{DH})}(r_2), \tag{52}$$

$$\lim_{r\to r_1^-} X_{\mathcal{E}}^{(\mathrm{DH})}(r) = \mathbb{1}, \qquad \lim_{r\to r_1^-} Y_{\mathcal{E}}^{(\mathrm{DH})}(r) = \Sigma_N^{(\mathrm{DH})}(r_1). \tag{53}$$

We define $\mathcal{E}_{\mathrm{DH}}(r_2)$ and $\mathcal{E}_{\mathrm{DH}}(r_1)$ by these one-sided limits. Each endpoint channel equals the corresponding additive-noise channel. If $r_1 = r_2$, then $\delta_1\delta_2 = 1$, the target POVM is rank one, and both channels reduce to the identity channel.

IV. GAUSSIAN-MODULATED COHERENT-STATE PROTOCOL

In the Gaussian-modulated coherent-state (GMCS) protocol, Alice samples the complex amplitude $\alpha = \alpha_1 + i\alpha_2$ from the Gaussian prior:

$$p_A(\alpha) = \frac{1}{2\pi V_{\mathrm{A}}}\exp\left[-\frac{|\alpha|^2}{2V_{\mathrm{A}}}\right]. \tag{54}$$

She then prepares the coherent state $|\alpha\rangle$. The states pass through the phase-insensitive Gaussian communication channel $\mathcal{N}_{A\to B}$, specified by $X_{\mathcal{N}} = \sqrt{T_{\mathrm{ch}}}\mathbb{1}$ and $Y_{\mathcal{N}} = (1 - T_{\mathrm{ch}} + \xi_{\mathrm{ch}})\mathbb{1}$, where T_{ch} is the channel transmittance and

$\xi_{\rm ch}$ is the output-referred excess-noise variance [34]. The receiver then uses homodyne or double-homodyne detection. The corresponding Alice-Bob mutual information is [12]

$$I_{\rm AB}^{\rm (H)} = \frac{1}{2}\log_2\left[1+\frac{4T_{\rm ch}V_{\rm A}}{\sigma_x+\xi_{\rm ch}}\right], \tag{55}$$

$$I_{\rm AB}^{\rm (DH)} = \frac{1}{2}\sum_{i=1}^{2}\log_2\left[1+\frac{4T_{\rm ch}V_{\rm A}}{2\sigma_i+\xi_{\rm ch}}\right], \tag{56}$$

where σ_x and σ_i are given by Eqs. (8) and (16), respectively. The superscripts H and DH label homodyne and double-homodyne detection throughout.

For the covariance-matrix calculations below, we work in the local quadrature basis obtained by applying $R(\phi)$ to Alice's coordinates and $R(-\phi)$ to Bob's coordinates. This basis change preserves the diagonal local blocks and the correlation block proportional to σ_z, while diagonalizing the target and environmental covariance matrices, and it does not change the symplectic eigenvalues or the information quantities. All covariance matrices below are written in this basis.

The covariance matrix of a two-mode squeezed-vacuum (TMSV) state after Bob's mode passes through the communication channel is [34]

$$\mathcal{I}_A\otimes\mathcal{N}_{A\to B}:\quad \Sigma_{\rm TMSV}=\begin{pmatrix} V\mathbb{1} & \sqrt{V^2-1}\sigma_z\\ \sqrt{V^2-1}\sigma_z & V\mathbb{1}\end{pmatrix}\mapsto \Sigma_{\rm AB}=\begin{pmatrix} V\mathbb{1} & c\sigma_z\\ c\sigma_z & V_{\rm B}\mathbb{1}\end{pmatrix}, \tag{57}$$

where $\sigma_z = {\rm diag}(1,-1)$, $\mathcal{I}_A$ is the identity channel acting on subsystem A, and the parameters of $\Sigma_{\rm AB}$ are

$$V = 1+4V_{\rm A},\quad c=\sqrt{T_{\rm ch}(V^2-1)}, \tag{58}$$

$$V_{\rm B} = T_{\rm ch}(V-1)+1+\xi_{\rm ch}. \tag{59}$$

We next apply either the additive-noise channel Φ_γ (33) or the attenuator channel $\mathcal{E}_\gamma$ (40) to Bob's mode of the TMSV state (57). For each channel, the corresponding reference POVM and outcome rescaling give the mutual information in Eqs. (55) and (56) because they reproduce the same receiver outcome statistics.

It follows that

$$\mathcal{I}_A\otimes\mathcal{E}_\gamma:\Sigma_{\rm AB}\mapsto\Sigma_{\rm AB}^{(\mathcal{E}_\gamma)}=\begin{pmatrix} V\mathbb{1} & c\cos\theta_\gamma\sigma_z\\ c\cos\theta_\gamma\sigma_z & \cos^2\theta_\gamma\left(V_{\rm B}\mathbb{1}+\tan^2\theta_\gamma\Sigma_{\rm env}^{(\gamma)}\right)\end{pmatrix}. \tag{60}$$

$$\mathcal{I}_A\otimes\Phi_\gamma:\Sigma_{\rm AB}\mapsto\Sigma_{\rm AB}^{(\Phi_\gamma)}=\begin{pmatrix} V\mathbb{1} & c\sigma_z\\ c\sigma_z & V_{\rm B}\mathbb{1}+\Sigma_N^{(\gamma)}\end{pmatrix},\quad \gamma\in\{{\rm H},{\rm DH}\}. \tag{61}$$

In the model with untrusted receiver noise, Eve holds a system E such that $\hat\rho_{\rm ABE}^{(\mathcal{C}_\gamma)}$ is a purification of

$$\hat\rho_{\rm AB}^{(\mathcal{C}_\gamma)} = (\mathcal{I}_A\otimes\mathcal{C}_\gamma)(\hat\rho_{\rm AB}). \tag{62}$$

Bob then applies the corresponding rank-one reference POVM. In $\chi_{\rm EB}$, the label B denotes Bob's rescaled classical outcome; in $\hat\rho_{\rm AB}$ and $\Sigma_{\rm AB}$, it denotes Bob's optical mode. Since the Alice-Bob-Eve state is pure before Bob's measurement and the reference POVM is rank one, the conditional Alice-Eve state is pure for every outcome. The Holevo information is therefore

$$\chi_{\rm EB}^{(\mathcal{C}_\gamma)} = S_{\rm E}-S_{\rm E|B} = S_{\rm AB}^{(\mathcal{C}_\gamma)}-S_{\rm A|B}^{(\mathcal{C}_\gamma)},\qquad \mathcal{C}_\gamma\in\{\Phi_\gamma,\mathcal{E}_\gamma\}. \tag{63}$$

The joint entropy

$$S_{\rm AB}^{(\mathcal{C}_\gamma)} = g(\nu_1^{(\mathcal{C}_\gamma)})+g(\nu_2^{(\mathcal{C}_\gamma)}),$$
$$g(\nu)=\frac{\nu+1}{2}\log_2\frac{\nu+1}{2}-\frac{\nu-1}{2}\log_2\frac{\nu-1}{2} \tag{64}$$

is determined by the symplectic eigenvalues of the covariance matrices in Eqs. (60) and (61):

$$\nu_{1,2}^{(\mathcal{C}_\gamma)} = \sqrt{\Delta_{\mathcal{C}_\gamma}/2\pm\sqrt{\Delta_{\mathcal{C}_\gamma}^2/4-D_{\mathcal{C}_\gamma}}},\quad D_{\mathcal{C}_\gamma}=\det\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}, \tag{65}$$

$$\Delta_{\mathcal{C}_\gamma} = \det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{11})+\det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{22})+2\det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{12}), \tag{66}$$

where $[\Sigma_{\mathrm{AB}}^{(\mathcal{C}_\gamma)}]_{11}$ and $[\Sigma_{\mathrm{AB}}^{(\mathcal{C}_\gamma)}]_{22}$ are the local covariance blocks, while $[\Sigma_{\mathrm{AB}}^{(\mathcal{C}_\gamma)}]_{12}$ is the correlation block.

The conditional entropy is

$$S_{\mathrm{A|B}}^{(\gamma)} = g(\nu_3^{(\gamma)}), \qquad \nu_3^{(\gamma)} = \sqrt{\det \Sigma_{\mathrm{A|B}}^{(\gamma)}}. \tag{67}$$

For a reference Gaussian POVM with covariance matrix $\Sigma_{\mathrm{id}}^{(\gamma)}$, the conditional covariance matrix obtained from the state after the channel is

$$\Sigma_{\mathrm{A|B}}^{(\mathcal{C}_\gamma)} = V\mathbb{1} - \Gamma_{\mathcal{C}_\gamma}\left(B_{\mathcal{C}_\gamma} + \Sigma_{\mathrm{id}}^{(\gamma)}\right)^{-1}\Gamma_{\mathcal{C}_\gamma}^{\mathrm{T}}, \tag{68}$$

where

$$\Gamma_{\mathcal{C}_\gamma} = \kappa_{\mathcal{C}}^{(\gamma)} c\sigma_z, \tag{69}$$
$$B_{\mathcal{C}_\gamma} = \left[\kappa_{\mathcal{C}}^{(\gamma)}\right]^2 V_{\mathrm{B}}\mathbb{1} + Y_{\mathcal{C}}^{(\gamma)}. \tag{70}$$

Equation (32) gives

$$B_{\mathcal{C}_\gamma} + \Sigma_{\mathrm{id}}^{(\gamma)} = \left[\kappa_{\mathcal{C}}^{(\gamma)}\right]^2 \left(V_{\mathrm{B}}\mathbb{1} + \Sigma_m^{(\gamma)}\right). \tag{71}$$

The factors $\kappa_{\mathcal{C}}^{(\gamma)}$ cancel in the Schur complement. The conditional covariance matrix is therefore independent of the channel choice and the squeezing parameter r:

$$\Sigma_{\mathrm{A|B}}^{(\gamma)} = V\mathbb{1} - c^2\sigma_z\left(V_{\mathrm{B}}\mathbb{1} + \Sigma_m^{(\gamma)}\right)^{-1}\sigma_z. \tag{72}$$

The corresponding symplectic eigenvalues are

$$\nu_3^{(\mathrm{H})} = \sqrt{V\left(V - \frac{c^2}{V_{\mathrm{B}} + \sigma_N}\right)}, \tag{73}$$
$$\nu_3^{(\mathrm{DH})} = V\sqrt{\frac{\left(V_{\mathrm{B}} + \delta_1 - \frac{c^2}{V}\right)\left(V_{\mathrm{B}} + \delta_2 - \frac{c^2}{V}\right)}{\left(V_{\mathrm{B}} + \delta_1\right)\left(V_{\mathrm{B}} + \delta_2\right)}}. \tag{74}$$

The asymptotic secret fraction is given by the Devetak-Winter formula [35]

$$K^{(\mathcal{C}_\gamma)} = \beta I_{\mathrm{AB}}^{(\gamma)} - \chi_{\mathrm{EB}}^{(\mathcal{C}_\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, \quad \mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}. \tag{75}$$

Here $\beta \in (0, 1]$ is the reverse-reconciliation efficiency.

For the additive-noise channel and the double-homodyne attenuator channel, we evaluate the symplectic eigenvalues using Eq. (65). The attenuator covariance matrix for homodyne has a simpler standard form, which gives the closed expression below.

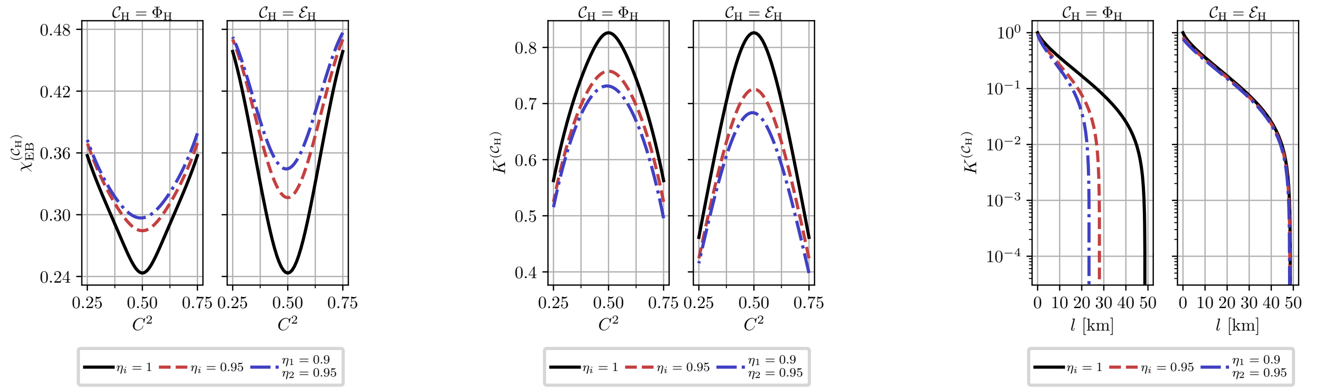
A. Homodyne detection

For the attenuator channel, Eqs. (60) and (46) give

$$\Sigma_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{H}})} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}\left(V_{\mathrm{B}} + \sigma_N\right)\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\mathrm{B}}\mathbb{1} \end{pmatrix}, \tag{76}$$
$$\tilde{V}_{\mathrm{B}} \equiv T(V-1) + 1 + \xi, \tag{77}$$
$$\tilde{c} \equiv \sqrt{T(V^2-1)}, \tag{78}$$

where $\tilde{V}_{\mathrm{B}}$ is Bob's scaled variance and $\tilde{c}$ is the scaled Alice-Bob correlation amplitude. The effective transmittance T and excess-noise variance ξ are

$$T = T_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}, \tag{79a}$$
$$\xi \equiv \xi_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}. \tag{79b}$$



(a) (b) (c)

Figure 2: (a) Holevo information, $\chi_{\rm EB}^{(\mathcal{C}_{\rm H})}$, and (b) asymptotic secret fraction, $K^{(\mathcal{C}_{\rm H})}$, as functions of the beam-splitter transmittance C^2, computed for the additive-noise and attenuator channels at the detector efficiencies specified in the subfigure titles. The parameters are $V=5$, $T_{\rm ch}=0.95$, $\xi_{\rm ch}=10^{-2}$, and $\beta=0.95$. (c) $K^{(\mathcal{C}_{\rm H})}$ versus channel length l for the two channels. The parameters are $V=5$, $\beta=0.95$, and $\xi_{\rm ch}=10^{-2}$; the fiber attenuation is 20 dB per 100 km.

Under the attenuator description of the homodyne receiver, the asymmetry-induced noise scales both the effective transmittance (79a) and the output-referred excess-noise variance (79b) by σ_x^{-1}.

The symplectic eigenvalues of $\Sigma_{\rm AB}^{(\mathcal{E}_{\rm H})}$ are [2]

$$\nu_{1,2}^{(\mathcal{E}_{\rm H})}=\frac{\omega_{\rm H}\pm\left(\tilde{V}_{\rm B}-V\right)}{2}, \tag{80}$$

$$\omega_{\rm H}=\sqrt{\left(V+\tilde{V}_{\rm B}\right)^2-4\tilde{c}^2}. \tag{81}$$

These eigenvalues are substituted into Eq. (64) to obtain the joint entropy $S_{\rm AB}^{(\mathcal{E}_{\rm H})}$.

Figure 2 shows the Holevo information and asymptotic secret fraction as functions of the receiver beam-splitter transmittance C^2, together with the asymptotic secret fraction as a function of channel length, for both channels. The attenuator scales the effective transmittance and excess-noise variance by the same factor (79). For the parameters shown, the attenuator channel gives higher Holevo information and a lower asymptotic secret fraction at $T_{\rm ch}=0.95$. In (c), its asymptotic secret fraction remains positive up to a greater channel length than for the additive-noise channel.

B. Double-homodyne detection

For the attenuator channel, Eqs. (60) and (50) give

$$\Sigma_{\rm AB}^{(\mathcal{E}_{\rm DH})}=\begin{pmatrix} V\mathbb{1} & c\cos\theta_{\rm DH}(r)\sigma_z & \\ c\cos\theta_{\rm DH}(r)\sigma_z & \begin{matrix}\cos^2\theta_{\rm DH}(r)\left(V_{\rm B}+\delta_1-\frac{e^{2r}}{\cos^2\theta_{\rm DH}(r)}\right) & 0\\ 0 & \cos^2\theta_{\rm DH}(r)\left(V_{\rm B}+\delta_2-\frac{e^{-2r}}{\cos^2\theta_{\rm DH}(r)}\right)\end{matrix}\end{pmatrix}. \tag{82}$$

The symplectic eigenvalues of this matrix can be calculated using invariants as in Eq. (65). These eigenvalues, and hence the joint entropy, depend explicitly on the squeezing parameter r, as in the additive-noise channel [12].

Figure 3 shows the Holevo information as a function of r for both channels. For ideal double-homodyne detection, $\delta_1=\delta_2=1$ and the admissible interval collapses to $r=0$ [12]. Both effective channels then reduce to the identity channel and give the same Holevo information, shown by the black dot.

When $\delta_1=\delta_2>1$, the Holevo information is symmetric about $r=0$ for both channels. It is maximal at $r=0$ for the additive-noise channel and minimal there for the attenuator channel. In the continuous extension defined below Eq. (51), the attenuator channel equals the corresponding additive-noise channel at $r=r_2$ and $r=r_1$. The Holevo

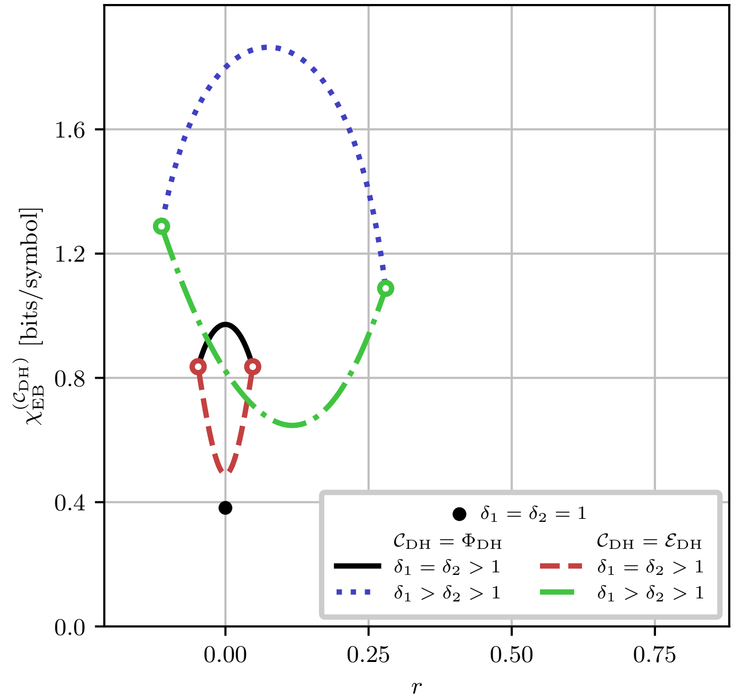


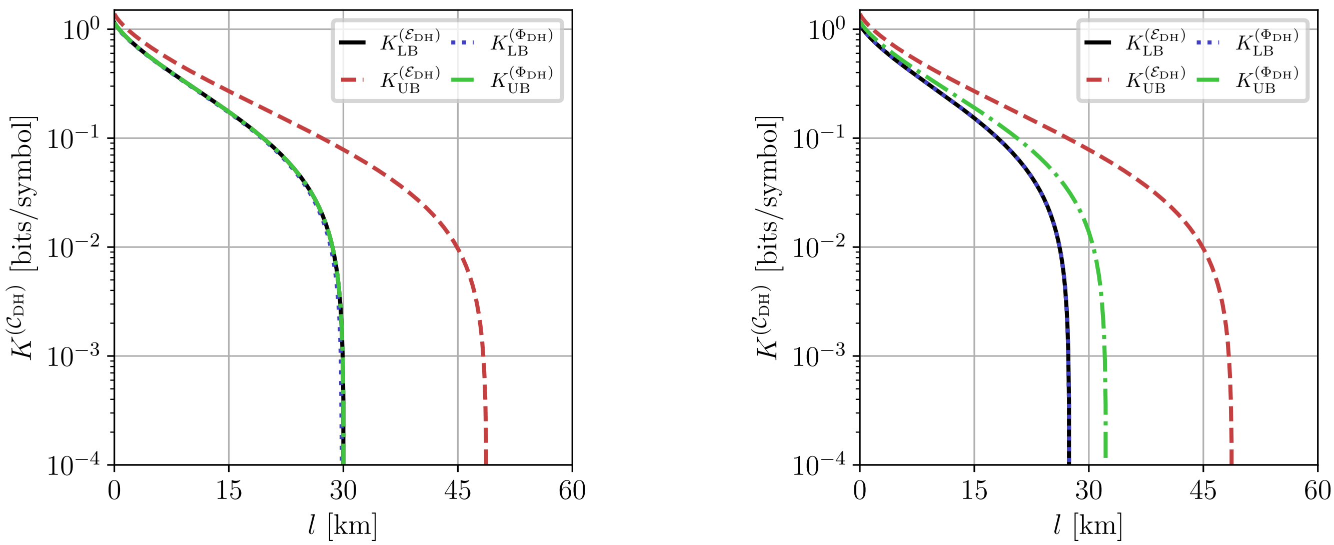
Figure 3: Holevo information, $\chi_{\mathrm{EB}}^{(\mathcal{C}_{\mathrm{DH}})}$, as a function of the squeezing parameter r for different values of δ_1 and δ_2. The black dot marks ideal double-homodyne detection. Solid and dash-dotted curves show the additive-noise channel; dashed and dotted curves show the attenuator channel. Color-matched white-filled dots mark the finite endpoint limits of the attenuator curves. At each endpoint, the limiting attenuator channel equals the corresponding additive-noise channel, although the environmental covariance matrix has no finite limit. The parameters are $\delta_1 = \delta_2 = 1.1$ for the solid and dashed curves and $\delta_1 = 1.75$, $\delta_2 = 1.25$ for the dash-dotted and dotted curves. The communication-channel parameters are $V = 5$, $T_{\mathrm{ch}} = 0.95$, and $\xi_{\mathrm{ch}} = 10^{-2}$.

information therefore agrees between the two channels at each endpoint, as shown by the color-matched white-filled dots in Fig. 3.

For the asymmetric parameter sets considered, where $\delta_1 \neq \delta_2$, neither curve is symmetric in r, and the joint entropies at the two endpoints $r = r_2$ and $r = r_1$ are unequal. For these parameter sets, the Holevo information is concave in r for the additive-noise channel and convex in r for the attenuator channel.

Figure 4 shows the ranges of the asymptotic secret fraction obtained by varying r over $[r_2, r_1]$ for both channels. For the parameter sets shown, convexity places the maximum of the continuously extended attenuator Holevo information at an endpoint, where the limiting channel equals the corresponding additive-noise channel. For the near-ideal value $\sigma_x^{(i)} = 1.05$, the additive-noise secret-fraction range is narrow, so the lower bounds for the two channels nearly coincide. In Fig. 4a, the channel lengths at which the upper and lower additive-noise bounds vanish differ by only 0.2 km. For both values of q shown, the attenuator range is wider than the additive-noise range. Increasing q from 1 to 1.25 makes the additive-noise range more pronounced.

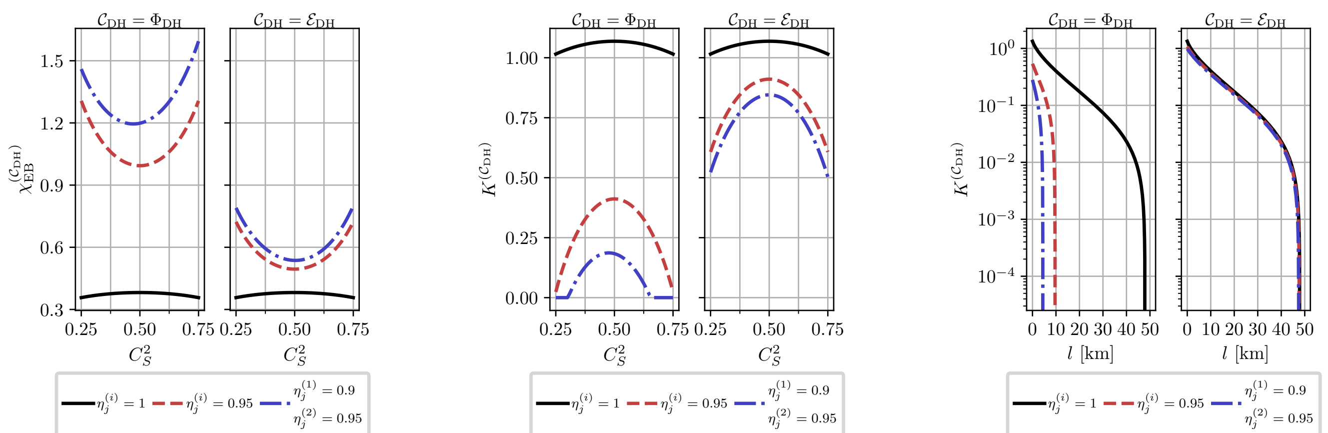
Figure 5 shows the Holevo information and asymptotic secret fraction as functions of the signal beam-splitter transmittance C_S^2, together with the asymptotic secret fraction as a function of channel length, for both channels. For the ideal receiver, $\eta_{1,2}^{(i)} = 1$ and $C_i^2 = S_i^2 = 1/2$ for $i \in \{1, 2\}$. These conditions give $\sigma_x^{(i)} = 1$ and $\delta_1\delta_2 = 1$ for every C_S^2. The target POVM is then rank one, both channels reduce to the identity channel, and the joint entropies coincide, as shown by the black curve. For the detector efficiencies below unity shown in the figure, the attenuator channel gives lower Holevo information than the additive-noise channel. The separation depends on the selected channel and the fixed value $r = (r_1 + r_2)/2$. For symmetric measurement noise, the midpoint is $r = 0$, which maximizes the additive-noise Holevo information and minimizes the attenuator Holevo information. For the parameters in (c), the attenuator secret fraction remains positive up to a greater channel length.



(a) $q = 1$

(b) $q = 1.25$

Figure 4: Asymptotic secret fraction $K^{(\mathcal{C}_{DH})}$ versus channel length l, computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq.(17)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for additive-noise and attenuator channel models (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\rm ch} = 10^{-2}$; the fiber loss is 20 dB per 100 km.



(a)

(b)

(c)

Figure 5: (a) Holevo information, $\chi_{\rm EB}^{(\mathcal{C}_{DH})}$, and (b) asymptotic secret fraction, $K^{(\mathcal{C}_{DH})}$, as functions of the signal beam-splitter transmittance C_S^2, computed for the additive-noise and attenuator channels at the detector efficiencies specified in the subfigure titles. The parameters are $V = 5$, $T_{\rm ch} = 0.95$, $\xi_{\rm ch} = 10^{-2}$, and $\beta = 0.95$. (c) $K^{(\mathcal{C}_{DH})}$ versus channel length l for the two channels. The parameters are $V = 5$, $\beta = 0.95$, and $\xi_{\rm ch} = 10^{-2}$; the fiber attenuation is 20 dB per 100 km. For both channels, $r = (r_2 + r_1)/2$. The beam splitters in the two homodyne branches are balanced, and detector efficiencies not specified in the subfigure titles are set to unity.

V. DISCUSSION AND CONCLUSION

In this paper, we constructed and compared two channel representations of the target homodyne and double-homodyne POVMs generated by the receiver imperfections. Each representation consists of a Gaussian channel, a rank-one reference POVM, and an invertible deterministic rescaling of the outcomes. Equation (32) relates the covariance matrices of the target and reference POVMs to the channel matrices. The receiver parameters determine the channel parameters; for double-homodyne detection, these parameters also depend on the squeezing parameter r.

The two choices reproduce the same conditional distribution of Bob's rescaled outcomes for every input state and therefore give the same Alice-Bob mutual information. They also give the same conditional covariance matrix in Eq. (72). Their Alice-Bob covariance matrices after the receiver channel generally differ. When the receiver noise is treated as untrusted, Eve purifies the corresponding state. The joint entropy and Holevo information can then depend on the selected channel even though Bob's observed statistics are fixed.

For homodyne detection, the attenuator environment is taken to be in the vacuum state, so the channel is a pure-loss channel. It scales the communication-channel transmittance and the output-referred excess-noise variance by σ_x^{-1}, as shown in Eq. (79). For the parameters in Fig. 2, the attenuator channel gives a larger Holevo information and a lower asymptotic secret fraction at high communication-channel transmittance. Its secret fraction remains positive over a greater channel length. The channel giving the lower secret fraction therefore changes with distance.

For double-homodyne detection, a fixed target POVM can be reproduced for a family of reference POVMs indexed by r. The receiver parameters fix $\Sigma_m^{(\mathrm{DH})}$, while r specifies the rank-one reference POVM. Complete positivity restricts r to the interval in Eq. (23). For the additive-noise channel, changing r redistributes the excess noise between the two quadratures according to Eq. (39). For the attenuator channel, it changes the attenuation coefficient and the covariance matrix of the pure Gaussian environmental input state according to Eqs. (50) and (51).

At each endpoint of a nondegenerate interval, one additive-noise quadrature variance vanishes. The environmental covariance matrix of the attenuator has no finite limit because one quadrature variance vanishes and the conjugate variance diverges. The channel itself has a finite one-sided limit and equals the corresponding additive-noise channel. If the interval collapses to one value, the noisy POVM is rank one and both channels reduce to the identity channel.

For the parameter sets in Fig. 3, the Holevo information is concave in r for the additive-noise channel and convex in r for the attenuator channel. With symmetric measurement noise, both curves are symmetric about $r = 0$. The coherent-state POVM at $r = 0$ maximizes the additive-noise Holevo information and minimizes the attenuator Holevo information. For the asymmetric parameter set, the extrema shift away from zero.

Figure 4 shows the secret-fraction ranges obtained by varying r over its admissible interval. For the near-ideal quadrature variances used there, the additive-noise range is narrow and the lower bounds for the two channels nearly coincide. The attenuator range is wider for both imbalance ratios shown. The additive-noise range becomes more pronounced as q increases from 1 to 1.25.

Figure 5 uses the midpoint $r = (r_1 + r_2)/2$. For symmetric measurement noise, this point maximizes the additive-noise Holevo information and minimizes the attenuator Holevo information. The separation between the two secret fractions in that figure therefore reflects the selected channel and the selected value of r.

The channel model specifies an assumption about the coupling of the optical mode to inaccessible degrees of freedom and Eve's access to them. A security estimate based on one channel is conditional on that assumption. If the physical receiver model does not select a unique compatible choice, a conservative bound within a specified class requires maximizing the Holevo information, or equivalently minimizing the secret fraction, over all channel representations in that class. The present comparison quantifies this dependence for two channel families. A bound independent of the channel choice would require an optimization over a specified class of Gaussian channels compatible with the target POVM.

The attenuator family considered here uses pure Gaussian environmental input states for $r \in (r_2, r_1)$, together with the endpoint channel limits. Mixed environmental input states and other Gaussian channels compatible with the same noisy POVM remain outside the present comparison.

Appendix A shows that independent pure losses in the two output arms of a lossless beam splitter can be absorbed into the detector efficiencies. A general passive lossy two-port transformation need not have attenuation that is diagonal in the output basis. Its quadrature gain, phase, and variance must then be calculated from the full complex scattering matrix.

Appendix B considers small classical amplitude fluctuations of a phase-locked local oscillator. In the linearized approximation, the additional variance generally depends on the input quadrature mean. Such fluctuations cannot, without a further approximation, be absorbed into the state-independent excess-noise covariance matrix used in the main text.

ACKNOWLEDGEMENTS

The work was supported by the Russian Science Foundation (project No. 24-11-00398).

Appendix A: Lossy beam splitter

In this appendix, we show that a lossless beam splitter followed by independent pure-loss channels in its two output arms has the scattering matrix $\mathbf{T}_{\rm BS} = \mathrm{diag}(\sqrt{\tau_1}, \sqrt{\tau_2})\mathbf{U}_{\rm BS}$.

First, consider the case of homodyne detection. The lossless unbalanced beam splitter mixes the input signal mode $\hat{a}$ and the local oscillator mode $\hat{a}_L$. This transformation is described by the unitary rotation matrix $\mathbf{U}_{\rm BS}$,

$$\begin{aligned}\begin{pmatrix}\hat{a}_1\\ \hat{a}_2\end{pmatrix} &= \mathbf{U}_{\rm BS}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix} = \begin{pmatrix}C & S\\ -S & C\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix}\\ &= \begin{pmatrix}C\hat{a}+S\hat{a}_L\\ -S\hat{a}+C\hat{a}_L\end{pmatrix},\end{aligned} \tag{A1}$$

where C and S are the amplitude transmission and reflection coefficients of the beam splitter BS, respectively.

We model the subsequent losses by passing each intermediate output mode $\hat{a}_{1,2}$ through a virtual beam splitter of transmittance $\tau_{1,2}$. This mixes each signal with an independent environmental vacuum mode $\hat{v}_{1,2}$, where $[\hat{v}_i, \hat{v}_j^\dagger] = \delta_{ij}$,

$$\begin{pmatrix}\hat{b}_1\\ \hat{c}_1\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_1} & -\sqrt{1-\tau_1}\\ \sqrt{1-\tau_1} & \sqrt{\tau_1}\end{pmatrix}\begin{pmatrix}\hat{a}_1\\ \hat{v}_1\end{pmatrix}, \tag{A2}$$

$$\begin{pmatrix}\hat{b}_2\\ \hat{c}_2\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_2} & -\sqrt{1-\tau_2}\\ \sqrt{1-\tau_2} & \sqrt{\tau_2}\end{pmatrix}\begin{pmatrix}\hat{a}_2\\ \hat{v}_2\end{pmatrix}. \tag{A3}$$

After tracing out the lost environmental modes $\hat{c}_{1,2}$, the retained output modes $\hat{b}_{1,2}$ satisfy

$$\begin{pmatrix}\hat{b}_1\\ \hat{b}_2\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_1}C & \sqrt{\tau_1}S\\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix} + \begin{pmatrix}-\sqrt{1-\tau_1}\hat{v}_1\\ -\sqrt{1-\tau_2}\hat{v}_2\end{pmatrix}. \tag{A4}$$

From Eq. (A4), the effective scattering matrix is

$$\mathbf{T}_{\rm BS} = \begin{pmatrix}\sqrt{\tau_1}C & \sqrt{\tau_1}S\\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C\end{pmatrix}. \tag{A5}$$

The associated noise operators are $\hat{F}_i = -\sqrt{1-\tau_i}\,\hat{v}_i$ for $i \in \{1,2\}$.

Preservation of the canonical commutation relations $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$ for the output fields requires

$$[\hat{F}_i, \hat{F}_j^\dagger] = \delta_{ij} - \sum_k T_{ik}T_{jk}^*, \tag{A6}$$

where T_{ik} are the elements of the scattering matrix $\mathbf{T}_{\rm BS}$ (A5). Evaluating this constraint explicitly using the elements of $\mathbf{T}_{\rm BS}$ yields:

$$[\hat{F}_1, \hat{F}_1^\dagger] = 1-\tau_1, \tag{A7}$$

$$[\hat{F}_2, \hat{F}_2^\dagger] = 1-\tau_2, \tag{A8}$$

$$[\hat{F}_1, \hat{F}_2^\dagger] = 0, \tag{A9}$$

which matches the commutators calculated from Eq. (A4).

We now relate this setup to the standard input-output description of passive lossy beam splitters [36, 37]. In this description,

$$\hat{\boldsymbol{b}} = \mathbf{T}_{\rm BS}\hat{\boldsymbol{a}} + \mathbf{A}\hat{\boldsymbol{v}}, \tag{A10}$$

where

$$\hat{\boldsymbol{a}} = \begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix}, \qquad \hat{\boldsymbol{b}} = \begin{pmatrix}\hat{b}_1\\ \hat{b}_2\end{pmatrix}, \qquad \hat{\boldsymbol{v}} = \begin{pmatrix}\hat{v}_1\\ \hat{v}_2\end{pmatrix}. \tag{A11}$$

Preservation of the canonical commutation relations requires

$$\mathbf{T}_{\rm BS}\mathbf{T}_{\rm BS}{}^\dagger + \mathbf{A}\mathbf{A}^\dagger = \mathbf{1}. \tag{A12}$$

For the scattering matrix in Eq. (A5),

$$\mathbf{T}_{\mathrm{BS}}\mathbf{T}_{\mathrm{BS}}{}^{\dagger} = \mathrm{diag}(\tau_1, \tau_2), \tag{A13}$$
$$\mathbf{A}\mathbf{A}^{\dagger} = \mathrm{diag}(1-\tau_1, 1-\tau_2). \tag{A14}$$

The scattering matrix therefore factors as

$$\begin{aligned}\mathbf{T}_{\mathrm{BS}} &= \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix} \begin{pmatrix} C & S \\ -S & C \end{pmatrix} \\ &= \sqrt{\mathbf{1} - \mathbf{A}\mathbf{A}^{\dagger}}\, \mathbf{U}_{\mathrm{BS}}. \end{aligned} \tag{A15}$$

Because $\mathbf{A}\mathbf{A}^{\dagger}$ is diagonal in the output basis, the two losses can be absorbed into the detector efficiencies as $\eta_i \to \tau_i \eta_i$ for $i \in \{1, 2\}$. The same rescaling applies in each branch of the double-homodyne receiver.

We finally consider a general passive lossy asymmetric beam splitter with scattering matrix [38]

$$\mathbf{S} = \begin{pmatrix} t & \rho \\ r_{\mathrm{BS}} & \tau \end{pmatrix}, \qquad \mathbf{S}\mathbf{S}^{\dagger} \preceq \mathbf{1}, \tag{A16}$$

where t and τ are complex transmission amplitudes and ρ and r_{BS} are complex reflection amplitudes. Such a device cannot be reduced to the lossless beam-splitter model by independent replacements of the two detector efficiencies.

For coherent signal and local-oscillator inputs, the detected output amplitudes are

$$|\alpha, \alpha_L\rangle \mapsto |\alpha_{\mathrm{out},1}, \alpha_{\mathrm{out},2}\rangle, \tag{A17}$$
$$\alpha_{\mathrm{out},1} = t\alpha + \rho\alpha_L, \tag{A18}$$
$$\alpha_{\mathrm{out},2} = r_{\mathrm{BS}}\alpha + \tau\alpha_L. \tag{A19}$$

We use the normalized Gaussian probability density

$$G(x; \sigma) \equiv \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{x^2}{2\sigma}\right). \tag{A20}$$

In the Gaussian approximation to the photon-count difference,

$$\mathsf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \tag{A21}$$

with

$$\begin{aligned}\sigma_G &= \eta_1|\alpha_{\mathrm{out},1}|^2 + \eta_2|\alpha_{\mathrm{out},2}|^2 \\ &= \left(\eta_1|\rho|^2 + \eta_2|\tau|^2\right)|\alpha_L|^2 \\ &\quad + \left(\eta_1|t|^2 + \eta_2|r_{\mathrm{BS}}|^2\right)|\alpha|^2 \\ &\quad + 2\,\mathrm{Re}\left[\left(\eta_1 t\rho^* + \eta_2 r_{\mathrm{BS}}\tau^*\right)\alpha\alpha_L^*\right], \end{aligned} \tag{A22}$$

$$\begin{aligned}\mu_G &= \eta_1|\alpha_{\mathrm{out},1}|^2 - \eta_2|\alpha_{\mathrm{out},2}|^2 \\ &= \left(\eta_1|\rho|^2 - \eta_2|\tau|^2\right)|\alpha_L|^2 \\ &\quad + \left(\eta_1|t|^2 - \eta_2|r_{\mathrm{BS}}|^2\right)|\alpha|^2 \\ &\quad + 2\,\mathrm{Re}\left[\left(\eta_1 t\rho^* - \eta_2 r_{\mathrm{BS}}\tau^*\right)\alpha\alpha_L^*\right]. \end{aligned} \tag{A23}$$

Let $\alpha_L = Le^{i\phi_L}$ and define

$$\Lambda_{\pm} = \eta_1|\rho|^2 \pm \eta_2|\tau|^2, \tag{A24}$$
$$\Gamma_{\pm} = \eta_1 t\rho^* \pm \eta_2 r_{\mathrm{BS}}\tau^*. \tag{A25}$$

At leading order in the strong local oscillator limit, the variance retains the term of order L^2, while the mean retains the offset of order L^2 and the signal term of order L:

$$\sigma_G \approx \Lambda_+ L^2, \tag{A26}$$
$$\mu_G \approx \Lambda_- L^2 + L|\Gamma_-|\langle\hat{x}_{\phi_{\mathrm{eff}}}\rangle, \tag{A27}$$
$$\phi_{\mathrm{eff}} = \phi_L - \arg\Gamma_-. \tag{A28}$$

Provided $\Gamma_- \neq 0$, the rescaled outcome and its variance are

$$x = \frac{\mu - \Lambda_- L^2}{L|\Gamma_-|}, \tag{A29}$$

$$\sigma_x^{(\mathrm{gen})} = \frac{\Lambda_+}{|\Gamma_-|^2}. \tag{A30}$$

Thus, a general passive lossy beam splitter still yields a quadrature measurement at leading order in L, but the quadrature phase, gain, and variance must be recalculated from its complex scattering amplitudes. The substitutions $\eta_i \to \tau_i \eta_i$ apply only to the diagonal output-loss model in Eq. (A5). Retaining the next term in σ_G introduces the signal-dependent correction

$$2L\,\mathrm{Re}\left(\Gamma_+ \alpha e^{-i\phi_L}\right), \tag{A31}$$

which lies beyond the strong local oscillator model with a state-independent measurement covariance used in the main text.

Appendix B: Fluctuating local oscillator

In this appendix, we extend the homodyne model to include classical amplitude fluctuations of a phase-locked local oscillator (LO).

Let $L = |\alpha_L|$ denote the mean LO amplitude. For fixed L, the photon-count-difference distribution in the strong-LO Gaussian approximation is [12]

$$\mathsf{P}_G(\mu \mid L) = G(\mu - \mu_G(L); \sigma_G(L)), \tag{B1}$$

$$\sigma_G(L) \approx (\eta_1 S^2 + \eta_2 C^2) L^2, \tag{B2}$$

$$\mu_G(L) \approx (\eta_1 S^2 - \eta_2 C^2) L^2 + CS(\eta_1 + \eta_2) L \langle \hat{x}_\phi \rangle. \tag{B3}$$

Here G is defined in Eq. (A20).

We write the instantaneous LO amplitude as $L + \Delta_L$, where Δ_L is a zero-mean Gaussian random variable:

$$p_L(\Delta_L) = G(\Delta_L; \sigma_L), \qquad \sqrt{\sigma_L} \ll L. \tag{B4}$$

Here σ_L is the variance of the LO amplitude fluctuations. Under the small-fluctuation condition, extending the integral over Δ_L to the full real line introduces a negligible tail with $L + \Delta_L < 0$.

The unconditional photon-count-difference distribution is the Gaussian mixture:

$$\mathsf{P}_{\mathrm{tot}}(\mu) = \int_{-\infty}^{\infty} \mathrm{d}\Delta_L \, G(\Delta_L; \sigma_L) \mathsf{P}_G(\mu \mid L + \Delta_L). \tag{B5}$$

We use a linearized Gaussian approximation: we set $\sigma_G(L + \Delta_L) \approx \sigma_G(L)$ and linearize the conditional mean,

$$\mu_G(L + \Delta_L) \approx \mu_G(L) + \mu_G'(L) \Delta_L, \tag{B6}$$

$$\mu_G'(L) = 2(\eta_1 S^2 - \eta_2 C^2) L + CS(\eta_1 + \eta_2) \langle \hat{x}_\phi \rangle. \tag{B7}$$

This approximation retains the broadening generated by the linearized conditional mean, freezes the conditional variance at $\sigma_G(L)$, and discards the quadratic term in $\mu_G(L + \Delta_L)$, including its contribution to the mean.

The Gaussian integral then gives

$$\mathsf{P}_{\mathrm{tot}}(\mu) \approx G\big(\mu - \mu_G(L); \sigma_G(L) + [\mu_G'(L)]^2 \sigma_L\big). \tag{B8}$$

The additional variance depends on $\langle \hat{x}_\phi \rangle$ through $\mu_G'(L)$. The linearized mixture therefore cannot, in general, be absorbed into the excess-noise variance σ_N or covariance matrix $\Sigma_m^{(\mathrm{H})}$ used in the main text.

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